

Customer Behavior Analysis

Prepared for Alfido Tech | Task 1 of 3 — Data Analytics Internship Program

Executive Summary

This report analyzes 250,000 e-commerce transactions from 49,673 customers (Jan 2020 – Sep 2023) to segment the customer base, understand purchase patterns, and identify churn risk. Customers were segmented using RFM (Recency, Frequency, Monetary) analysis, cross-validated with K-Means clustering. Revenue is essentially flat over the full period (~\$15M/month with no growth trend), and is heavily concentrated: the top two segments (**Champions** and **Loyal Customers**, 37.6% of customers) generate 52.5% of total revenue. A large **Lost** segment (18.8% of customers, ~580 days since last purchase) represents a recoverable revenue pool. The dataset's pre-labeled Churn field was tested and found to carry no measurable behavioral signal (churn rate is a flat ~20% across every segment, category, and demographic cut) — recency-based engagement is used as the practical churn-risk indicator instead. Five actionable recommendations are provided in Section 6.

Transactions	Unique Customers	Total Revenue	Recorded Churn Rate	RFM Segments
250,000	49,673	\$681.3M	20.0%	7

Key Findings

- Revenue is flat, not growing — monthly revenue has hovered around \$15M with no upward trend across 3.5+ years.
- Revenue is concentrated in a minority of customers — Champions + Loyal Customers are 37.6% of the base but generate 52.5% of revenue.
- A large “Lost” segment (18.8% of customers) has gone quiet (avg. 580 days since last purchase) but still represents 8.9% of historical revenue — a recoverable pool.
- Clothing and Books drive the most orders, but Electronics and Home generate meaningfully higher revenue per order — a cross-sell opportunity.
- The provided Churn label carries no behavioral signal in this dataset (flat ~20% everywhere) — recency/engagement should be used as the practical churn-risk indicator instead.

Top 5 Recommendations (see Section 6 for full detail)

1. Launch a win-back campaign for “Lost” and “At Risk (High Value)” segments.
2. Build a loyalty/VIP program for Champions and Loyal Customers to protect concentrated revenue.
3. Convert “New/Promising” customers to repeat buyers with a structured onboarding offer.
4. Cross-sell Electronics/Home to high-frequency Clothing/Books buyers to lift order value.
5. Instrument a real, time-based churn definition (e.g. 90+ days inactive) for future tracking.

1. Introduction & Data Overview

Dataset: Customer Behavior Analysis (Kaggle, bhanupratapbiswas) — *ecommerce_customer_data_custom_ratios.csv*. The file contains transaction-level e-commerce records: one row per purchase, with customer demographics, product category, payment method, return flag, and a pre-labeled churn indicator.

Column	Description
Customer ID	Unique customer identifier
Purchase Date	Timestamp of the transaction
Product Category	Electronics, Home, Clothing, or Books
Product Price / Quantity	Unit price and units purchased
Total Purchase Amount	Transaction value (used as the revenue field)
Payment Method	Credit Card, PayPal, Cash, or Crypto
Returns	1 if the item was returned, else 0 (missing → assumed 0)
Age / Gender / Customer Name	Customer demographics
Churn	Pre-labeled churn flag (1 = churned) — see Section 5 for a critical assessment

Scope: 250,000 transactions, 49,673 unique customers, spanning 2020-01-01 to 2023-09-15.

2. Data Cleaning & Methodology

2.1 Cleaning Steps

- **Duplicate column:** “Age” and “Customer Age” were identical for all 250,000 rows → “Customer Age” dropped.
- **Missing values:** “Returns” was missing for 19.0% of rows; filled with 0 (assumed “not returned”). This is a stated assumption, not a confirmed business rule.
- **Duplicate rows:** checked, none found.
- **Invalid values:** checked for non-positive price, quantity, or purchase amount; none found.
- **Date parsing:** “Purchase Date” converted to datetime for time-based aggregation.
- **Data quality flag:** “Total Purchase Amount” does not equal “Product Price × Quantity” for 99.98% of rows — a known characteristic of this dataset variant. “Total Purchase Amount” is used as the authoritative revenue field throughout this analysis since it best represents realized transaction value.

2.2 Feature Engineering

Transactions were aggregated to one row per customer, computing Recency (days since last purchase, relative to one day after the dataset's last date), Frequency (transaction count), Monetary (total spend), Average Order Value, Tenure (days between first and last purchase), Return Rate, favorite category/payment method, and demographics.

2.3 Segmentation Methodology

RFM scoring: Recency, Frequency, and Monetary were each split into quintiles and scored 1 (worst) to 5 (best). Combinations of these scores were mapped to seven interpretable business segments (Champions, Loyal Customers, New/Promising, At Risk (High Value), Needs Attention, Lost, Others).

K-Means clustering (cross-check): standardized Recency/Frequency/Monetary features were also clustered with K-Means (k tested from 2–8, k=4 selected for interpretability near the elbow point). The resulting clusters (High-Value, Steady/Mid-Value, Low-Engagement, Dormant/Low-Value) closely mirror the rule-based RFM tiers, which gives confidence that the segmentation reflects real structure in the data rather than arbitrary thresholds.

3. Customer Segments & Profiles

Segment	Customers	% of Cust.	Avg. Recency (days)	Avg. Frequency	Avg. Monetary	% of Revenue
Champions	8,739	17.6%	61	7.65	\$21,785	27.9%
Loyal Customers	9,957	20.0%	134	6.07	\$16,787	24.5%
Others	8,651	17.4%	366	4.81	\$12,316	15.6%
At Risk (High Value)	3,759	7.6%	382	7.08	\$20,455	11.3%
Lost	9,339	18.8%	580	2.59	\$6,488	8.9%
New / Promising	5,403	10.9%	68	3.24	\$8,802	7.0%
Needs Attention	3,825	7.7%	190	3.36	\$8,426	4.7%

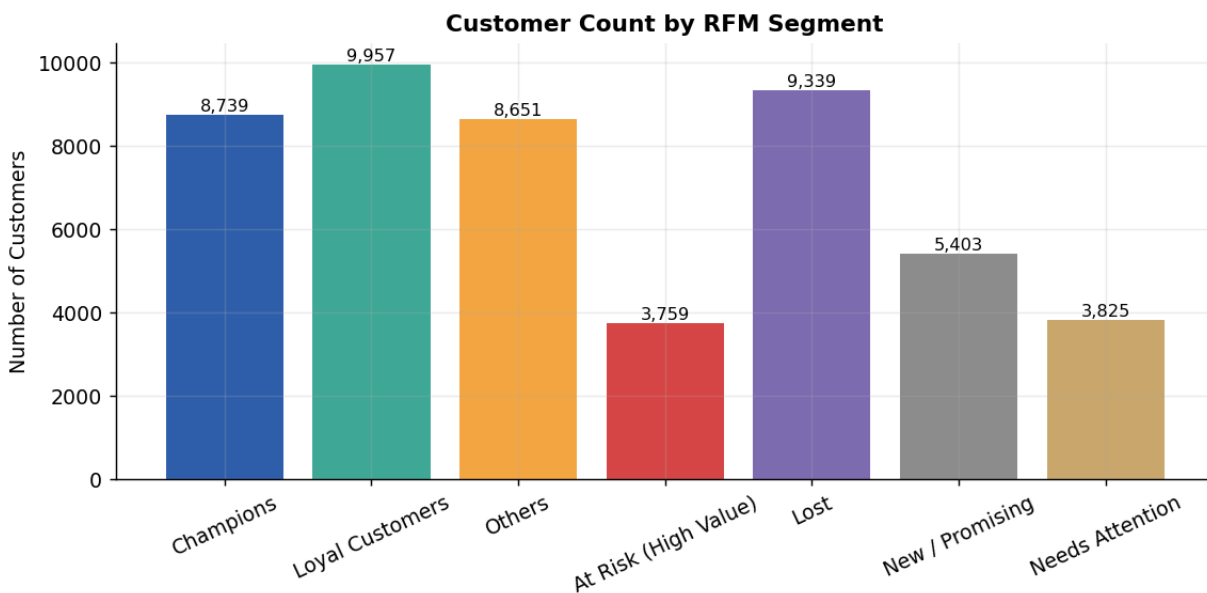


Figure 1. Customer count by RFM segment.

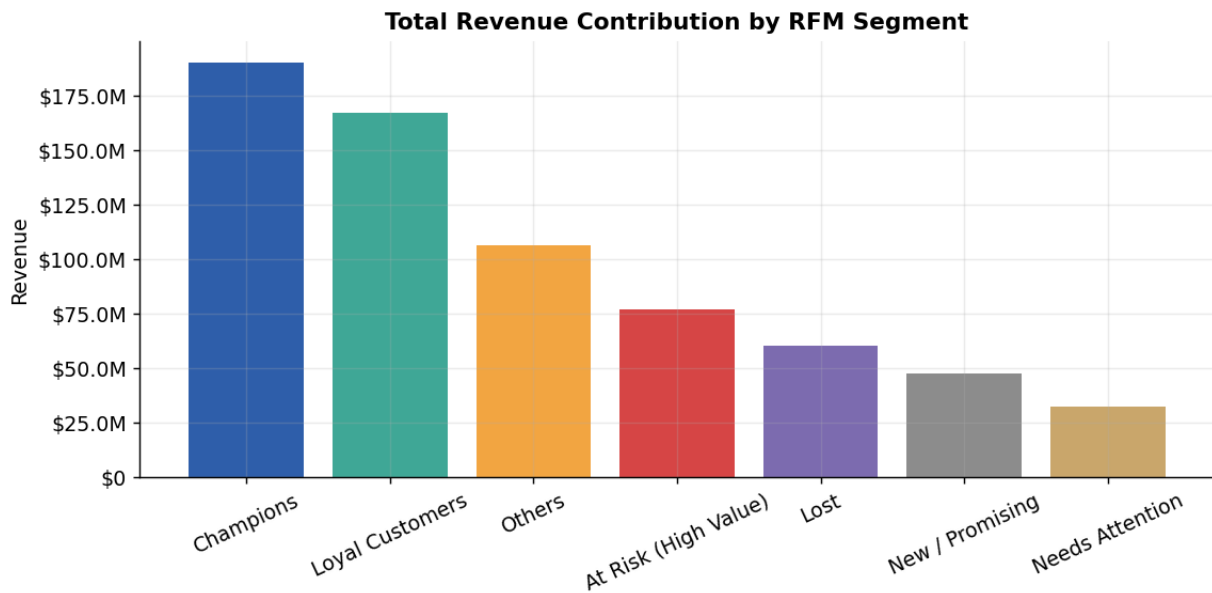


Figure 2. Total revenue contribution by RFM segment. Revenue is clearly top-heavy: Champions and Loyal Customers together account for over half of all revenue despite being well under half the customer base.

Segment Descriptions

Champions: Recent, frequent, high-spend customers. Alfido Tech's most valuable and most loyal group.

Loyal Customers: Solid recency/frequency/monetary scores across the board — dependable repeat buyers.

At Risk (High Value): Historically high spenders who have gone quiet (avg. 382 days) — the highest-priority group to re-engage given their spend history.

New / Promising: Recent first-time or early-stage buyers with low frequency so far — the door is open, but a second purchase hasn't happened yet.

Needs Attention: Middling across all three RFM dimensions — neither strongly engaged nor clearly lost.

Lost: Long inactive (avg. 580 days), low historical frequency and spend — the largest segment by count, and the lowest priority per-customer, but sizeable in aggregate.

Others: Customers who don't cleanly fit the above rule combinations — a mixed-profile residual group.

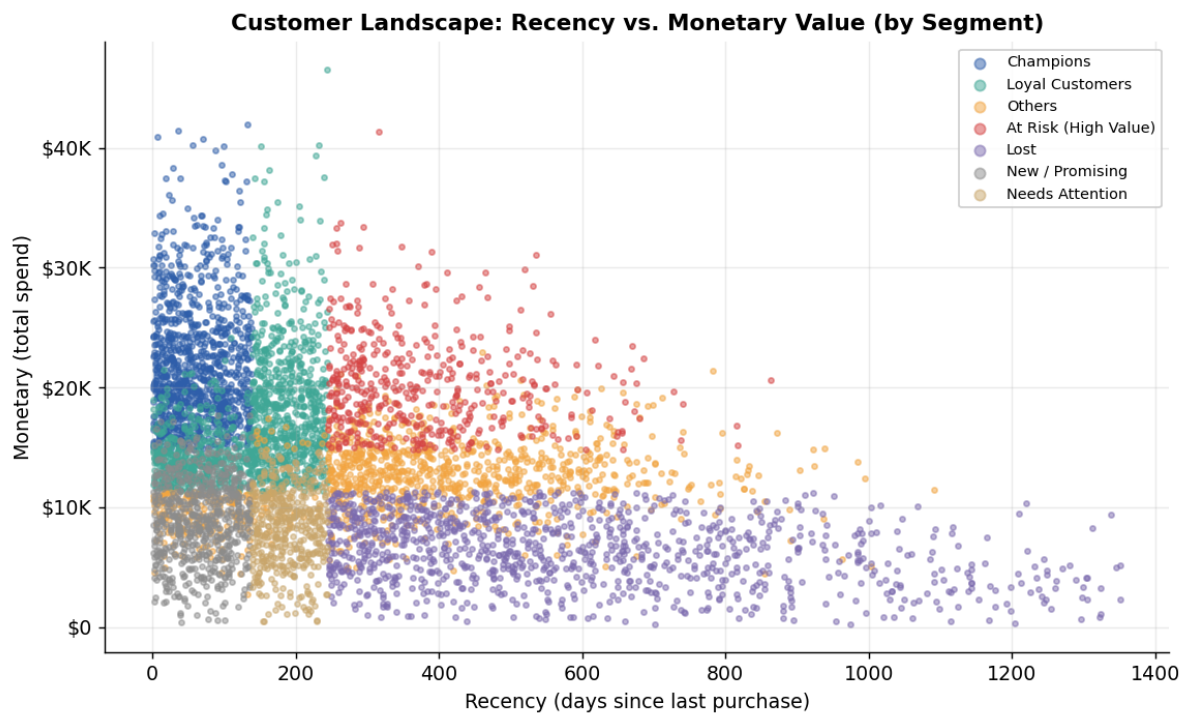


Figure 3. Recency vs. Monetary value, colored by segment (6,000-customer sample). The segmentation cleanly separates the customer base along both engagement recency and spend.

4. Purchase Patterns & Retention Trends

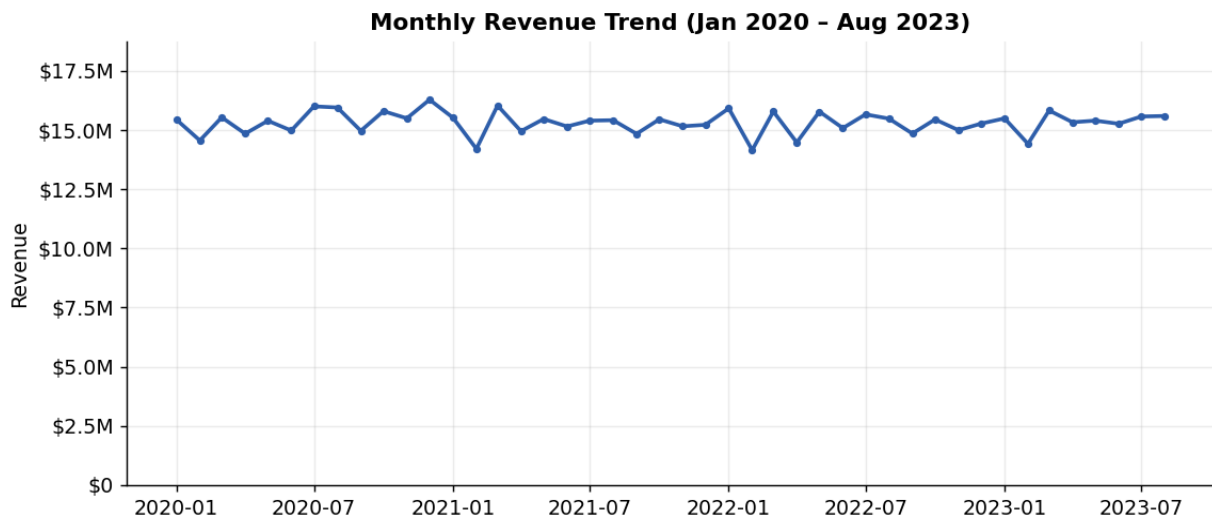


Figure 4. Monthly revenue, Jan 2020 – Aug 2023 (final partial month excluded). Revenue is essentially flat around \$15M/month — no growth trend and no clear seasonality.

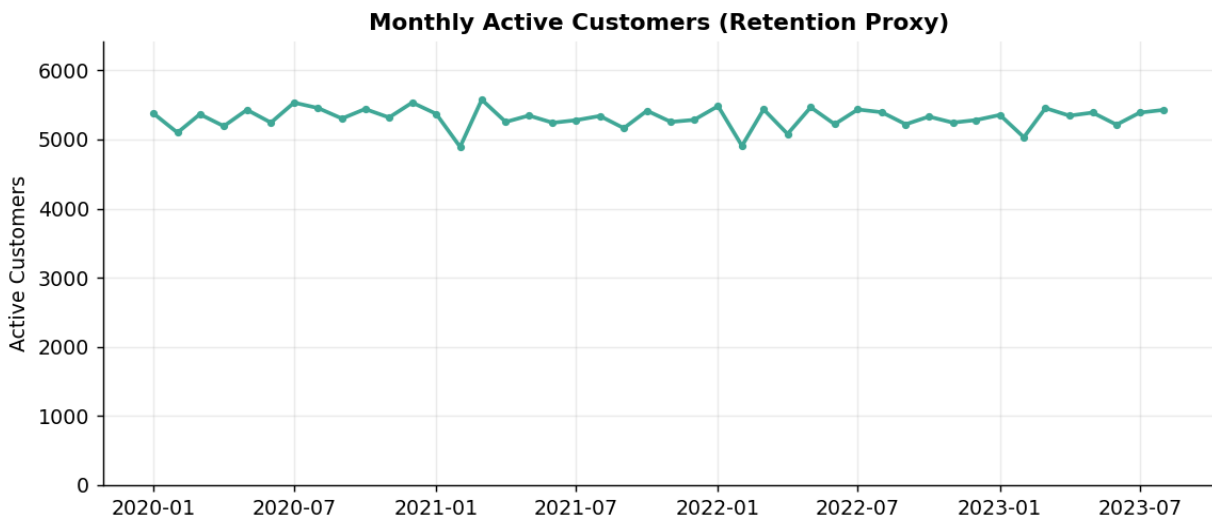


Figure 5. Monthly active customers, used as a retention proxy. Mirrors the flat revenue pattern — the active customer base is neither growing nor shrinking materially month to month.

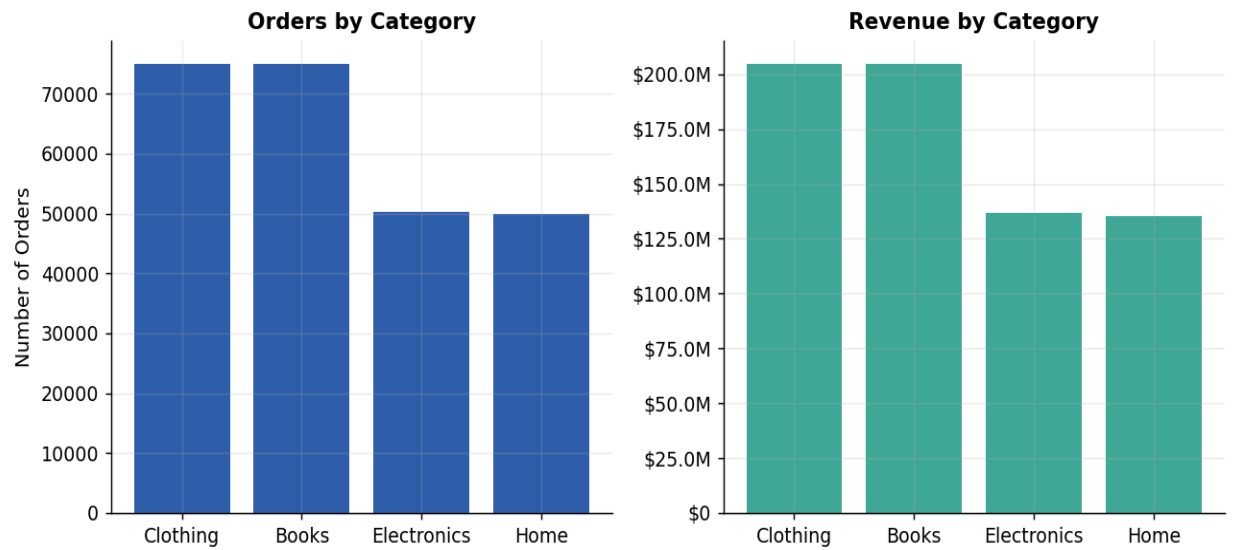


Figure 6. Orders vs. revenue by product category. Clothing and Books lead in order volume, but Electronics and Home earn more per order on average.

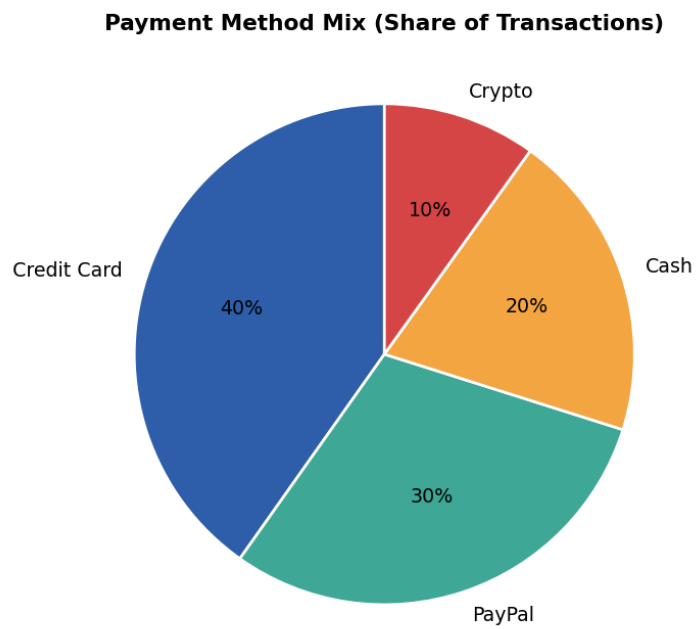


Figure 7. Payment method mix. Credit Card leads (40%), followed by PayPal (30%), Cash (20%), and Crypto (10%).

5. Churn Risk Analysis

The dataset includes a pre-labeled Churn field (1 = churned, 20.0% overall). Before using this field to drive recommendations, we tested whether it correlates with any observed customer behavior.

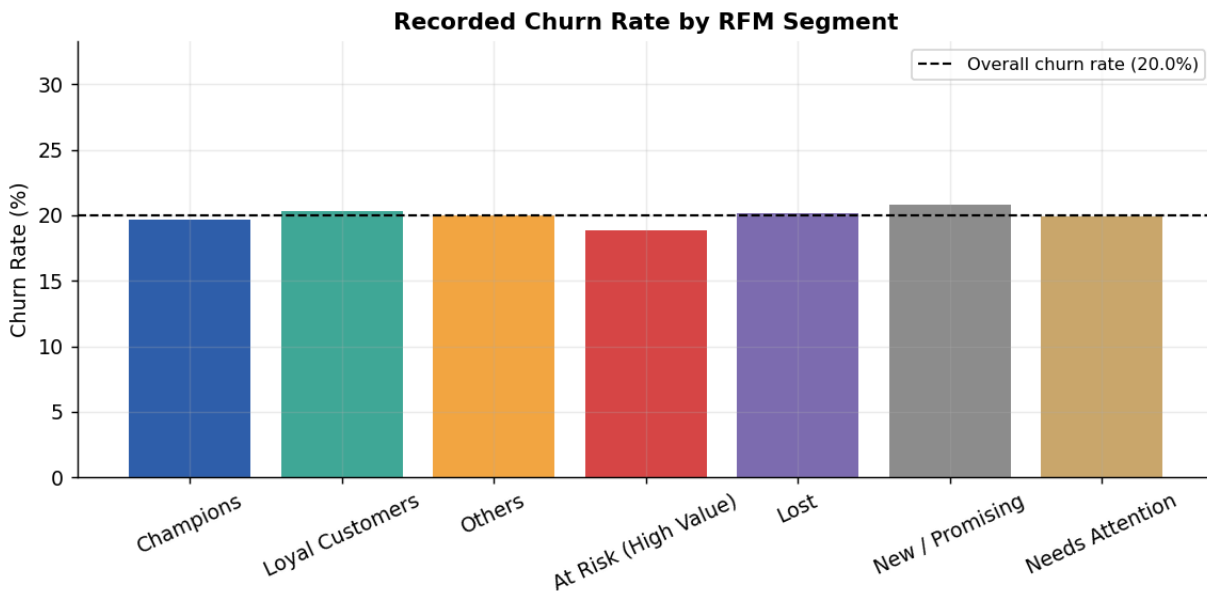


Figure 8. Recorded churn rate by RFM segment. Rates are flat (18.9%–20.8%) across every segment, with no meaningful gap between “Champions” and “Lost.”

Correlation of the Churn flag with Recency, Frequency, Monetary, Average Order Value, Tenure, Return Rate, and Age was checked directly — all coefficients were below $|r| = 0.01$. Churn rate was also flat across Gender (19.7% vs. 20.3%), product category (19.6%–21.0%), and payment method (19.9%–20.8%).

Conclusion: the Churn field in this dataset does not carry a behaviorally-grounded signal — a known characteristic of this synthetic dataset variant, and an important validation step before building any retention strategy on top of it. Rather than treat the label as ground truth, this report uses **Recency and RFM segment membership** (particularly “At Risk” and “Lost”) as the practical, defensible proxy for churn risk — consistent with how most retention teams operate before they have a validated churn outcome to model against.

6. Actionable Recommendations for Alfido Tech

1. Win back “At Risk (High Value)” and “Lost” customers

Prioritize the 3,759 “At Risk (High Value)” customers first — they still represent 11.3% of total revenue despite ~382 days of inactivity. Run targeted win-back offers (personalized discount, reminder of past purchase categories) via email/SMS. The larger “Lost” segment (9,339 customers, 8.9% of revenue) can be addressed with a lower-cost, broader re-engagement campaign.

2. Build a loyalty/VIP program for Champions and Loyal Customers

These two segments are 37.6% of customers but 52.5% of revenue — protecting and growing this base has outsized impact. Tiered perks, early access to new products, or free shipping thresholds can increase frequency further and reduce the risk of them drifting toward “At Risk.”

3. Convert “New/Promising” customers into repeat buyers

This segment has strong recency but low frequency (avg. 3.2 orders) — the relationship is new, not yet proven. A structured onboarding flow (e.g., a second-purchase incentive within 30 days, a welcome series highlighting complementary categories) can convert first-time buyers into habitual ones.

4. Cross-sell Electronics/Home to high-frequency Clothing/Books buyers

Clothing and Books drive the most orders but Electronics and Home earn more revenue per order. Bundle promotions, “customers like you also bought” placements, and targeted category cross-promotion can lift average order value across the largest-volume customer base.

5. Instrument a real, time-based churn definition going forward

Because the existing Churn field shows no behavioral signal, define and track an operational churn rule (e.g., “no purchase in 90+ days”) starting now. This gives Alfido Tech a churn label that actually reflects behavior, enabling future predictive modeling and a way to validate whether recommendations 1–4 are working.

Prepared as part of the Alfido Tech Data Analytics Internship Program — Task 1: Customer Behavior Analysis. Full code, methodology, and additional exploratory visualizations are available in the accompanying Jupyter notebook (Customer_Behavior_Analysis.ipynb).